Music Genre Classification

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1. Introduction

The main goal of this project was to build a machine learning system that can automatically classify music into different genres. Music genre classification is widely used today in music streaming platforms, recommendation systems, playlist generation, and organization of personal music libraries. Even though deep learning models have become popular for audio tasks, classical machine learning techniques can still produce strong results when meaningful audio features are extracted from each track.

In this project, I used the GTZAN dataset, which contains ten different genres such as classical, metal, pop, jazz, and rock. I began by extracting detailed audio features that capture the sound and structure of each clip. I then trained several supervised learning models to compare how well they could classify music using these engineered features. I also explored unsupervised clustering and simple generative models to gain a deeper understanding of how the dataset behaves without labels. Overall, the project aimed to evaluate different learning methods, understand their strengths and limitations, and identify opportunities for future improvement.

2. Problem Statement, Setup, and Methods

The core problem addressed in this project is predicting the genre of a music track based solely on its audio characteristics. To do this, I worked with the GTZAN dataset, which originally includes 1000 tracks across ten genres. One audio file was corrupted and could not be processed, so the final dataset contained 999 tracks. Each clip is approximately thirty seconds long. I assumed that these clips are representative enough to capture the essence of each genre and that engineered features would be sufficient for traditional machine learning models to learn meaningful patterns.

To prepare the data, I extracted a variety of audio features using the Librosa library in Python. These features included Mel-Frequency Cepstral Coefficients (MFCCs), chroma features, spectral contrast, tonnetz features, and several spectral statistics such as centroid, bandwidth, rolloff, and RMS energy. For each feature, the mean and standard deviation were computed across time, resulting in a fixed-length feature vector of 157 values for each track. This allowed the audio signals to be represented in a way that machine learning models could process effectively.

For supervised learning, I trained logistic regression, k-nearest neighbors, linear SVM, RBF-kernel SVM, decision trees, random forests, and AdaBoost. Hyperparameter tuning was done using GridSearchCV with five-fold cross-validation. The dataset was split into eighty percent for training and twenty percent for testing, using stratification to maintain balanced genre distributions.

I also evaluated unsupervised learning models, including K-Means clustering and Bayesian Gaussian Mixture Models, to see whether different genres naturally form distinguishable groups without labels. Finally, I explored simple generative models, such as Gaussian Naive Bayes and class-conditional Gaussian Mixture Models, to understand how well these models could capture the structure of each genre.

3. Results

Among all the supervised models, the Random Forest classifier performed the best. It achieved a test accuracy of 78.5 percent and a macro F1-score of 0.779. Logistic regression and the SVM models also performed fairly well, with accuracies around 71 to 73 percent. k-Nearest neighbors achieved around 64 percent, while the decision tree and AdaBoost models performed significantly worse. The strong performance of the Random Forest model suggests that it is effective at capturing nonlinear patterns and handling large sets of mixed features.

The unsupervised models did not perform well at separating genres. K-Means clustering achieved a normalized mutual information score of only 0.160 and an adjusted Rand index of 0.083. The Bayesian Gaussian Mixture Model produced almost identical results. These low scores show that, without labels, the data does not naturally fall into clusters that match human-defined music genres. This makes sense because many genres share instruments, rhythms, and production styles, meaning that their boundaries are not clear-cut.

The generative models showed interesting behavior. Gaussian Naive Bayes achieved a resubstitution accuracy of about 48 percent, which is expected given its strong independence assumptions. Class-conditional Gaussian Mixture Models achieved nearly perfect resubstitution accuracy, close to 99.9 percent. This result indicates heavy overfitting: the model is very good at memorizing the training data but may not generalize well to new audio samples. Together, these results show that classical generative models either oversimplify the data or overfit it.

4. Insights and Future Work

This project revealed several insights about classical machine learning methods for music genre classification. First, engineered audio features such as MFCCs and spectral statistics are powerful enough to support strong classification performance. The Random Forest model demonstrated that traditional techniques can still be effective, even when compared to the deep learning approaches commonly used today. Second, the difficulty of the unsupervised task highlights the fact that genre is a human-defined label rather than a naturally occurring grouping in audio data. Genres often blend into each other, making it hard for clustering algorithms to separate them without explicit guidance.

There are several directions for future work. One major improvement would be switching from engineered features to mel-spectrogram images and training a convolutional neural network, which is the modern standard for audio classification. Data augmentation techniques such as pitch shifting, time stretching, and adding background noise could help improve model robustness. Using larger and more diverse datasets would also likely improve generalization. Finally, exploring sequence-based models such as CRNNs or transformers could provide deeper insight into temporal structure and genre-specific musical patterns.

5. Conclusion

This project successfully built a full machine learning pipeline for music genre classification using classical audio features and supervised models. The Random Forest model achieved the strongest performance with 78.5 percent accuracy, showing that traditional machine learning is still highly competitive for audio analysis tasks. Unsupervised models and generative models provided additional perspective and highlighted both the complexity of the dataset and the limitations of simpler modeling approaches. Overall, the project demonstrates the effectiveness of feature engineering in audio classification and provides a strong foundation for future work using deep learning methods.

References

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